

Marcos Arnold

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EDUCATION

Colorado College

Colorado Springs, CO

Bachelor of Arts, Double Major in Computer Science and Sociology

May 2027

- Awards: Crown-Goodman Presidential Scholarship, Keller Family Venture Grant, HSF Scholar, ColorStack

EXPERIENCE

Human-Computer Interaction (HCI) Research Assistant

November 2025 – Present

Colorado College Department of Computer Science

Colorado Springs, CO

- Identified a dimensional-accuracy gap in Meshy AI's text-to-image-to-3D pipeline while prototyping adaptive grips, tracing the failure to lost measurement instructions across 60+ generation and print iterations, and scoping a prompt-level fix to preserve dimensional accuracy from user text input through 3D print.
- Synthesized 15+ ACM papers on intellectual and developmental disabilities into co-design requirements for generative 3D prototyping, shaping a testing plan across transformer-based models with Macalester College.

Software Development Intern

May 2025 – October 2025

Colorado College Office of Information Technology

Colorado Springs, CO

- Shipped a Django/HTMX system centralizing 23 institutional applications across a five-tier organizational hierarchy, improving cross-department visibility and reducing reporting time by an estimated 40%.
- Architected a PostgreSQL schema modeling many-to-many relationships across divisions, service areas, business processes, departments, and applications to reflect the institution's actual administrative structure.

PROJECTS

Retention | 3rd Place, Best Use of TokenRouter, UC Berkeley AI Hackathon 2026

- Engineered the generation layer of a multi-agent video pipeline, translating LLM-generated creative direction into structured Pika prompts and Deepgram TTS specs, by parameterizing style, pacing, visual complexity, and motion controls into the generation call, enabling fully automated text-to-video regeneration.
- Closed a 5-iteration generator-evaluator loop, debugging agent communication in Band while re-scoring each AI-generated video with Meta's TRIBE v2 brain-encoding model, driving a ~60% increase in viewer engagement.

Milkr | Python, React, TypeScript, FastAPI, Plaid, Anthropic Claude

- Built a Claude-powered catalog pipeline scraping issuer pages and newsroom announcements into structured SQLite rate data, running weekly diff checks and quarterly bonus scans to keep rewards current automatically.
- Engineered a two-engine ranking system computing per-transaction expected value across merchant categories, rotating bonuses, caps, and redemption multipliers, recommending cards where rewards outweigh annual fees.

Streetwise | Python, FastAPI, PRAW, Leaflet.js, APScheduler, Anthropic Claude

- Launched a Chicagoland mobility alert map with a confidence-scoring pipeline corroborating Reddit, CTA, and Metra events within a 500m/30-min geo-temporal window, filtering low-confidence reports to surface disruptions.
- Deployed an LLM-based extraction layer to convert unstructured transit alerts and social posts into structured, geocoded events, powering a live Leaflet map that surfaces civic disruptions as they emerge.

LEADERSHIP EXPERIENCE

Co-Director

February 2025 – Present

Colorado College Coding Club

Colorado Springs, CO

- Design and lead bi-weekly workshops for 20+ students, building curriculum around community interest and professional goals across data structures, web development, and career preparation.
- Founded Tiger HaCCs, Colorado College's first interdisciplinary, civic hackathon, securing \$3K funding from the City of Colorado Springs, with all teams later invited to pitch at Exponential Impact, a startup accelerator.

Webmaster Lead

September 2025 – Present

The Sounds of Colorado College (SOCC) Radio

Colorado Springs, CO

- Expanded audio and archival content discoverability for 500+ weekly users by building automated DJ scheduling workflows, integrating live radio streaming, and digitizing zine collections into interactive web flipbooks.